Year 9 · Chapter 10 · Surface area and volume: answers (teacher copy)

10.6 Circumference and arc length

(booklet p. 1)

WE DO examples (teacher)

1. Ex 1a: $C = 2 \times \pi \times 5 = 31.4$ cm
2. Ex 1b: $C = \pi \times 12 = 37.7$ m
3. Ex 1c: $P = \frac{1}{2} \times \pi \times 10 + 10 = 25.7$ cm
4. Ex 1d: $P = \frac{1}{4} \times 2 \times \pi \times 6 + 6 + 6 = 21.4$ m
5. Ex 2a: $l = 60/360 \times 2 \times \pi \times 9 = 9.4$ cm
6. Ex 2b: $P = 9.42\ldots + 9 + 9 = 27.4$ cm
7. Ex 2c: $P = 10 + 6 + 10 + \frac{1}{2} \times \pi \times 6 = 35.4$ m (not the dashed side)
8. Ex 2d: $50 = 2 \times \pi \times r$, so $r = 50 \div (2 \times \pi) = 8.0$ cm

Easy

1. 8 cm
2. 5 m
3. 13 cm
4. 4.5 m
5. 18.8 cm
6. 31.4 m
7. 7.9 m
8. 44.0 cm
9. 12.6 cm
10. 50.3 m
11. 9.4 m
12. 25.1 cm
13. 28.3 m
14. 75.4 mm
15. 20.6 cm
16. 30.8 m
17. 25.7 m
18. 14.3 cm
19. 35.7 m
20. 21.4 cm
21. 22.0 m
22. 51.4 cm
23. 28.6 m

Medium

1. 6.3 m
2. 15 m² (10.5)
3. 7.9 cm
4. 24.6 cm
5. 28.6 cm
6. 25.9 cm
7. 4.45 m to 4.55 m (10.2)
8. 3.0 m
9. 2.5% (10.3)

Worded problems

1. $C = \pi \times 3.2 = 10.05$ m; $10.05 \times \$12 = \$121$
2. $C = 2 \times \pi \times 33 = 207.3$ cm; $100\,000 \div 207.3 = 482.3$, so 482 full turns
3. $84 + 84 + \pi \times 64 = 369.1$, so 369 m